

# The bottleneck was never the learning.

Automated biobank-to-OMOP harmonisation for federated analysis

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Raw biobank data in. Harmonised OMOP out. Quality controlled.  
Trained across sites. Nothing leaves the institution.

## THE PROBLEM

# Federated learning has worked for years. It still almost never happens.

## WHAT WE ASSUME IS HARD

# The learning

Solved and shipped. NVFlare, FedAvg, secure aggregation.

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Not a privacy problem. A language problem.

## WHAT STOPS IT

# The data

- Every biobank stores its data differently.
- A hand-built ETL per source takes months.
- No common model until somebody builds one. Per site. By hand.

# We automated the step everyone did by hand



Raw files in at one end, a trained model at the other.  
The only thing that crosses a boundary is a weight vector.

# An agent wrote the mapping. It matches the expert ETL.

254,882

of 254,891 measurement values  
identical to OHDSI ETL-SYNTHETA

0

differences in diagnoses,  
on every start date

8 s

to convert 2.1 GB of raw  
data, no server

11

person rows differ, and the  
contract is the one that is right

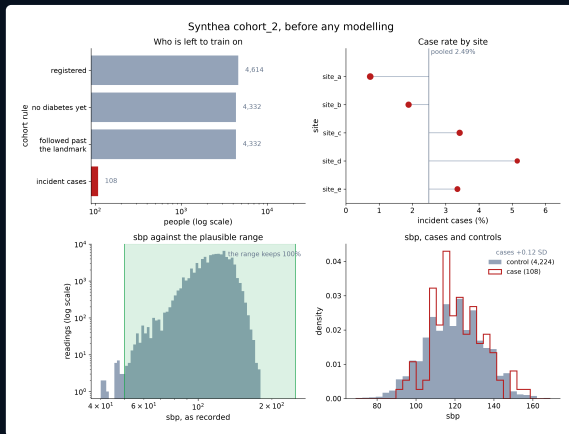
**Blind run.** The mapping was written from the contract and a profile of the raw files, not from an OMOP output to copy.

**QC caught what both conversions let through.** Nine LDL values below zero. HbA1c with a median of 3.9 %.

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Months of ETL, in eight seconds. And checkable.

# QC answers the question before the model does



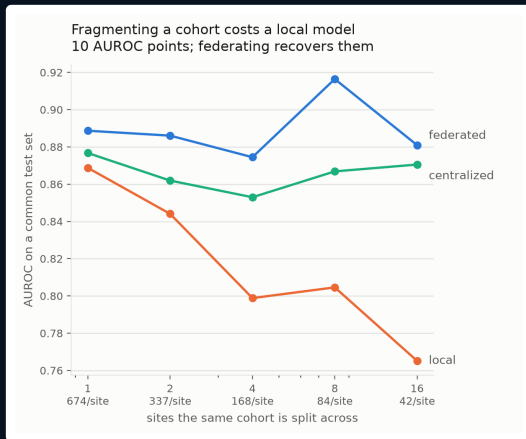
Federation pays off only when the sites differ.

|                            | Synthea | UK Biobank |
|----------------------------|---------|------------|
| Case rate across sites     | 7.0×    | 1.6×       |
| Worst site vs. pooled      | 0.79 SD | 0.03 SD    |
| Span of site BMI medians   | 4.90    | 0.28       |
| SBP in the plausible range | 100 %   | 94 %       |

Eight centres within 0.03 SD of the pooled mean are eight copies of one cohort.

One cohort has something to federate. The other does not.  
The figure says so before you train.

# Federation recovers what fragmentation costs



- One cohort of 964 patients, split across 1 to 16 sites.
- Alone, a site falls from **0.87 to 0.77**.
- Federated, it holds at **0.87–0.92**, level with pooled.
- Nobody shared a row.

**At scale:** 1,000,000 patients, 3.39 billion rows. Scan 7.9 s, feature matrix 7.7 s, one epoch 1.1 s, 6.2 GB. A real NVFlare job, three clients: **42 s for five rounds**.

**Collaboration stops being a technical problem.  
It becomes a choice.**

BACKUP

# Appendix

Method, the second route through the vocabulary, the negative control, the full benchmarks, and the limits.

# Two routes into OMOP, one contract out

## ROUTE A — THE CODES ARE IN THE CONTRACT

- **Profile.** A script reports which columns exist, what is empty, which codes are frequent.
- **Write the recipe.** The agent writes a YAML mapping:  $M \rightarrow 8507$ ,  $F \rightarrow 8532$ , LOINC 39156-5  $\rightarrow$  concept 3038553.
- **A person approves the recipe.**
- **Cook.** DuckDB executes it and writes the four tables. Same input, same output.

Synthea. 8 s for 2.1 GB, no server, no code change per source.

## ROUTE B — THE CODES NEED TRANSLATING

- **Prepare** the extract from the raw files.
- **Propose.** For each source code, the scripts search the local Athena vocabulary and write candidate standard concepts into a review list.
- **A person decides, code by code:** which target, on what evidence. The agent explains; it never approves.
- **Apply.** Only approved codes become OMOP rows. The rest stay visible as `concept_id = 0`.

UK Biobank ICD-10. All eight assessment centres pass contract validation.

# What the QC gate checks

## CONTRACT CHECK

- Column names, order, types
- YYYY-MM-DD dates
- Only permitted concept IDs
- Unique IDs
- Every `person_id` present in `person`

## QUALITY REPORT

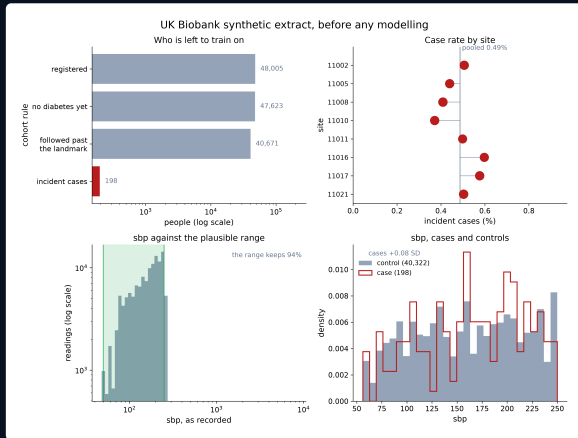
- Missingness per column
- Date logic: index before follow-up, events inside the observation period
- Units, and unknown units
- Mapping rate, share of records on `concept_id = 0`
- The ten most frequent unmapped codes

## PLAUSIBILITY

- Lab values against a plausible range
- BMI outside 10–80 kg/m<sup>2</sup>
- Systolic BP outside 60–250 mmHg
- Optional comparison against a published reference conversion

**A failed check is triaged by whose fault it is.** If the recipe is wrong, it gets fixed, at most three attempts. If the data is wrong — negative LDL, an implausible HbA1c — that is reported, not bent into the recipe. Every site runs this on its own output, locally, before training.

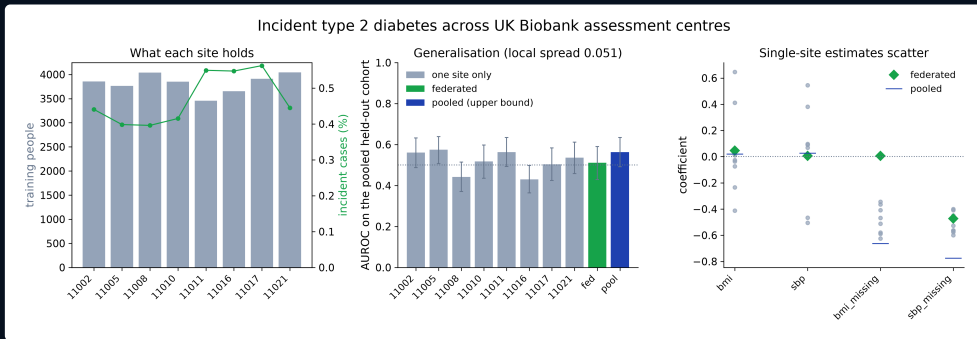
# UK Biobank: the honest null result



- UK Biobank draws every synthetic field independently, so no model beats chance. Every AUROC interval covers 0.5.
- 40,671 people, eight assessment centres, 198 incident cases.
- Single sites still report signal: the eight score 0.430 to 0.575, federated scores 0.511.
- Their BMI coefficients run from  $-0.41$  to  $+0.65$ , against a federated 0.05.

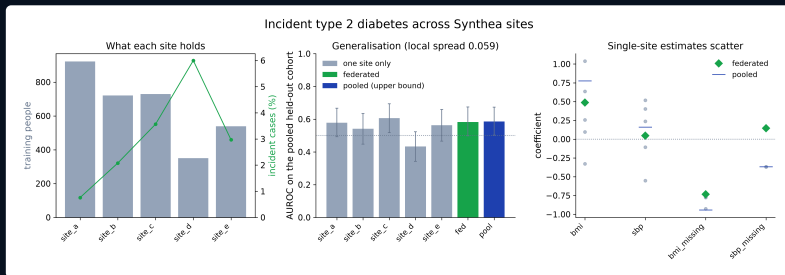
A small site will hand you a plausible-looking model built on nothing. The federated estimate, and the QC figure before it, both say so.

# The same pipeline on data with no signal



Middle panel: every bootstrap interval covers 0.5. Right panel: single-site coefficients scatter around the federated and pooled estimates.

# Five Synthea sites that differ

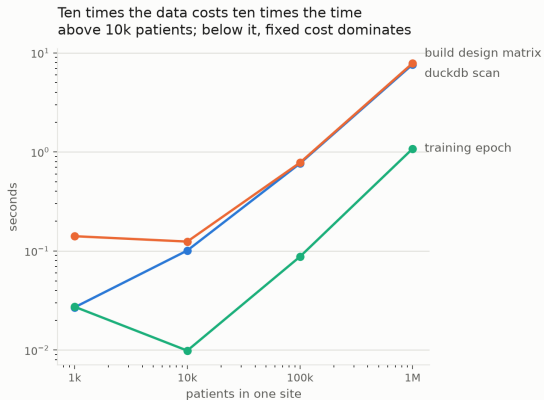


| model                    | AUROC        | 95 % CI     |
|--------------------------|--------------|-------------|
| site_d alone, 350 people | 0.433        | 0.343–0.523 |
| site_b alone             | 0.541        | 0.447–0.635 |
| site_e alone             | 0.563        | 0.465–0.659 |
| site_a alone             | 0.578        | 0.493–0.666 |
| site_c alone             | 0.606        | 0.518–0.694 |
| <b>federated</b>         | <b>0.582</b> | 0.499–0.675 |
| pooled (upper bound)     | 0.586        | 0.503–0.674 |

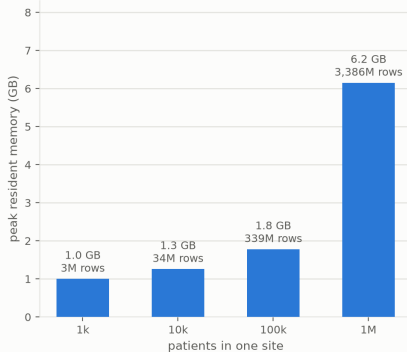
FedAvg recovers the pooled model without any site sharing a row. The smallest site would have done worse than chance on its own.

With 23 positives held out the intervals overlap heavily. They support *federated matches pooled and beats the weakest site*, not a significant pairwise gap, and we do not claim one.

# A million patients, 3.4 billion rows



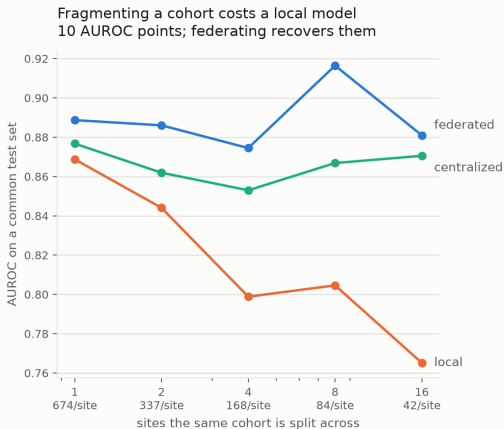
1000x the rows costs 6x the memory, since the raw rows are never materialised



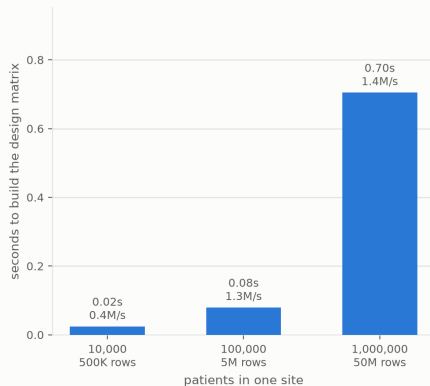
A real NVFlare FedAvg job over the same 1,000,000 patients and 3.39B rows, split across 3 clients: 42s for 5 rounds, 8.4s per round.

MIMIC's real row density, replicated upward. The raw rows are never materialised, so 1000x the rows costs 6x the memory.

# The full federation benchmark



Design matrix build time stays sub-second to 50M rows; the scan never leaves DuckDB



Left: the slide-6 panel in full. Right: design matrix build time stays sub-second to 50 million rows, because the scan never leaves DuckDB.

# What this is, and what it is not

## WHAT IT DOES

- Turns raw source CSVs into the four contract tables, reproducibly and locally, with no database server.
- Takes a new source with a new YAML file and no code change.
- Checks any output folder against the contract, including somebody else's.
- Profiles raw data so a mapping starts from facts.
- Runs FedAvg over the result through NVFlare, with the OMOP dataloader.

## WHAT IT DOES NOT DO YET

- Prove itself on a source family beyond Synthea and UK Biobank.
- Look up concepts on its own. The IDs come from the contract, checked against Athena once, by a person.
- Cover more than four tables. No vocabulary tables, visits, drugs or procedures.
- Run unattended. An AI drafts every mapping; a person approves every mapping.

**Proof of concept.** The Synthea and UK Biobank data are synthetic, the sites are simulated, and the federated benchmark uses one real cohort. Next: HUNT cloud over AWS — real infrastructure, a real institutional boundary.

# Where everything lives

`omop_skill/`

Subproject 1. Raw source data to the contract's OMOP tables, and the checks on them.

`omop_skill/SKILL.md`

The instructions any coding agent follows to run the conversion.

`omop_skill/review/`

Route B for UK Biobank: Athena proposals, human approval, visual QC.

`src/omopflare/`

Subproject 2. OMOP features and dataloaders for NVFlare.

`examples/ukb_diabetes/`

Raw UK Biobank → OMOP → federated diabetes risk, end to end.

`examples/synthea_diabetes/`

The same pipeline on a cohort that has signal.

`contract/data_contract.md`

The contract both subprojects hand over on.

`omop_skill/RESULTS.md`

Every number on slide 4, and how it was tested.

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**Concordia** · Project 5, collaborativebioinformatics hackathon, Copenhagen 2026

charles · solvi · lukas · maria · max · nik · mia · kev